JAVA CORE PROGRAMMING

Comprehensive 100-Question Multiple Choice Examination

Total Questions: 100
Topics: OOP, Variables, Controls & I/O

SECTION 1: CLASSES & OBJECTS (Q1 – Q10)

1. Which keyword is used to create an instance of a class in Java?

A) `class`
B) `new`
C) `object`
D) `create`

2. What is a class in Java?

A) A primitive data type
B) A blueprint/template for objects
C) An instance of memory
D) A method execution block

3. Which memory area stores objects created using the `new` operator?

A) Stack Memory
B) Heap Memory
C) Method Area
D) PC Register

4. What is the default value of an uninitialized object reference inside a class?

A) `0`
B) `""`
C) `null`
D) `undefined`

5. How many objects of a class can be created in a single program?

A) Only one
B) Up to 256
C) Unlimited (subject to memory)
D) Defined by class modifier

6. Which of the following best describes an object's state?

A) Its defined methods
B) Values of its attributes/fields
C) Its access modifier
D) Its memory address

7. Which of the following is TRUE about anonymous objects?

A) They have no name
B) Created without `new`
C) Cannot call methods
D) Stored permanently

8. What happens when an object no longer has any references pointing to it?

A) Compiler error occurs
B) Eligible for Garbage Collection
C) Program terminates immediately
D) Automatically cloned

9. Which operator is used to access members of a class through an object?

A) `->`
B) `.` (Dot operator)
C) `::`
D) `#`

10. Can a Java file contain multiple non-public classes?

A) Yes
B) No
C) Only if they are static
D) Only inside main method

SECTION 2: CONSTRUCTORS (Q11 – Q20)

11. What is the return type of a Java constructor?

A) `void`
B) `int`
C) Same class type
D) Constructors have no return type

12. When is a default constructor provided by the compiler?

A) Always
B) When no constructors are defined
C) When class is public
D) When class is abstract

13. Which keyword calls a constructor from another constructor in the same class?

A) `super()`
B) `this()`
C) `extent()`
D) `new()`

14. Where must `this()` or `super()` be placed inside a constructor?

A) Anywhere
B) First statement only
C) Last statement
D) Inside a try block

15. Can a constructor be marked as `final`?

A) Yes
B) No
C) Only if private
D) Only in abstract classes

16. What is Constructor Overloading?

A) Same parameters, different name
B) Multiple constructors with different parameters
C) Overriding parent constructor
D) Making a constructor private

17. Which constructor is called when executing `Demo d = new Demo(5);`?

A) Default constructor
B) Parameterized constructor
C) Copy constructor
D) Static constructor

18. What happens if a class defines only a parameterized constructor and you call `new Demo()`?

A) Runs successfully
B) Compilation error
C) Runtime exception
D) Returns null

19. Can a constructor be declared `private`?

A) No, causes syntax error
B) Yes, prevents external instantiation
C) Yes, but cannot run code
D) Only in interfaces

20. What is the primary purpose of a constructor?

A) Destroy objects
B) Initialize instance variables
C) Allocate heap space only
D) Import packages

SECTION 3: ENCAPSULATION (Q21 – Q30)

21. What is Encapsulation in Java?

A) Inheriting properties from parent
B) Wrapping data and methods into a single unit
C) Executing multiple forms of methods
D) Creating multiple classes in one file

22. How do you achieve full encapsulation in a class?

A) Make variables public
B) Make variables private & provide getters/setters
C) Make methods private
D) Extend Java's Object class

23. Which access modifier restricts variable access to its own class only?

A) `protected`
B) `public`
C) `default`
D) `private`

24. What is a read-only encapsulated class?

A) Class with only setter methods
B) Class with only getter methods
C) Class with private methods
D) Class declared as final

25. What is the conventional return type of a Setter method for a field `age`?

A) `int`
B) `boolean`
C) `void`
D) `String`

26. Which of the following is a key benefit of encapsulation?

A) Increased execution speed
B) Data hiding and control over validation
C) Automatic memory cleanup
D) Prevents compilation errors

27. What happens if a public getter method returns a reference to a mutable private object?

A) Encapsulation may be compromised
B) Compiler error
C) Runtime NullPointerException
D) Memory leak

28. Which component is encapsulated in a typical Java Bean class?

A) Only static variables
B) Private fields with public getters/setters
C) Public fields only
D) Abstract methods

29. What access modifier is applied when no keyword is specified for a field?

A) `private`
B) `protected`
C) Package-private (default)
D) `public`

30. Can getter and setter methods be declared as `static` for instance fields?

A) Yes
B) No, instance fields require non-static methods
C) Only getter methods
D) Only setter methods

SECTION 4: METHODS & METHOD OVERLOADING (Q31 – Q40)

31. What defines a method signature in Java?

A) Return type + Method name
B) Method name + Parameter list
C) Access modifier + Name
D) Exceptions + Return type

32. Can Method Overloading be achieved by changing ONLY the return type?

A) Yes
B) No, causes compilation error
C) Yes, at runtime
D) Only if return type is void

33. Which keyword is used to pass variable arguments to a method?

A) `args...`
B) `varargs`
C) Ellipsis `...`
D) Array brackets `[]`

34. What is the return type of a method that does not return any value?

A) `null`
B) `void`
C) `empty`
D) `static`

35. Can a `static` method access non-static instance fields directly?

A) Yes
B) No, requires an object reference
C) Yes, if in same package
D) Yes, if fields are public

36. What is method overriding?

A) Same method in same class
B) Subclass redefining superclass method
C) Calling private method
D) Changing parameter count

37. Which keyword prevents a method from being overridden?

A) `static`
B) `const`
C) `final`
D) `abstract`

38. What parameter list does `public static void main` accept?

A) `String args`
B) `String[] args`
C) `Object[] args`
D) `int[] args`

39. What happens if a method calls itself endlessly without a base condition?

A) OutOfMemoryError
B) StackOverflowError
C) InfiniteLoopException
D) NullPointerException

40. Can abstract methods have a body inside an abstract class?

A) Yes
B) No
C) Only if static
D) Only if final

SECTION 5: LOCAL & INSTANCE/GLOBAL VARIABLES (Q41 – Q50)

41. Where are local variables created in memory?

A) Heap Memory
B) Stack Memory
C) Method Area
D) Code Segment

42. What is the default value of an uninitialized local variable of type `int`?

A) `0`
B) `null`
C) No default value (causes error)
D) Garbage value

43. Which modifier CANNOT be applied to a local variable?

A) `final`
B) `public`
C) Both A and B
D) None of the above

44. Instance variables are declared inside a class but outside of what?

A) Package
B) Any method, constructor, or block
C) Imports
D) Interface

45. What is the lifetime of a static (class) variable?

A) Until method finishes
B) As long as the class is loaded
C) Until object is dereferenced
D) 5 seconds

46. What happens when a local variable has the same name as an instance variable?

A) Compiler Error
B) Variable Shadowing
C) Instance variable overrides local
D) Garbage collection triggers

47. How do you access a shadowed instance variable `x` inside a method?

A) `this.x`
B) `super.x`
C) `x.this`
D) `local.x`

48. Where are static variables stored in memory?

A) Call Stack
B) Metaspace / Method Area
C) Register
D) Heap Root

49. What is the default value of an uninitialized instance variable of type `boolean`?

A) `true`
B) `false`
C) `0`
D) `null`

50. Which keyword is used to declare a constant variable in Java?

A) `const`
B) `constant`
C) `final`
D) `static`

SECTION 6: DECISION MAKING (IF-ELSE & SWITCH CASE) (Q51 – Q60)

51. What must be the evaluation type of the condition inside an `if()` statement?

A) `int` (0 or 1)
B) `boolean`
C) Any numeric type
D) `String`

52. Which of the following primitive types CANNOT be used in a standard `switch` expression?

A) `int`
B) `char`
C) `double`
D) `byte`

53. What occurs if a `switch` case lacks a `break` statement?

A) Compilation error
B) Fall-through to subsequent cases
C) Program halts immediately
D) Default case is skipped

54. Is the `default` case mandatory inside a `switch` block?

A) Yes
B) No
C) Only if cases are missing
D) Only for String switches

55. What is the result of `(5 > 3) ? "Yes" : "No"`?

A) `"Yes"` B) `"No"`
C) `true` D) Compilation Error

56. Which operator represents Logical AND in an `if` condition?

A) `&` B) `&&`
C) `AND` D) `|`

57. Can duplicate `case` values exist within the same `switch` block?

A) Yes B) No, results in compile error
C) Yes, if different types D) Only in enum switches

58. Starting from Java 7, which object type can be used in a `switch` statement?

A) `StringBuffer` B) `String`
C) `Double` D) `ArrayList`

59. What executes if no `case` matches and no `default` is provided in a switch?

A) Exception thrown B) Nothing executes; exits switch
C) First case runs D) Infinite loop

60. What is the output of `if(false); else System.out.print("OK");`?

A) Prints nothing B) Prints `OK`
C) Compile Error D) Runtime Error

SECTION 7: LOOPS & ITERATIONS (Q61 – Q75)

61. Which loop is guaranteed to execute its body at least once?

A) `for` loop B) `while` loop
C) `do-while` loop D) Enhanced `for` loop

62. What is the output of `for(int i=0; i<3; i++);` after execution?

A) `i = 0` B) Loop runs 3 times doing nothing
C) Infinite loop D) Compile error

63. Which statement immediately terminates the innermost loop?

A) `continue` B) `exit`
C) `break` D) `skip`

64. What does the `continue` statement do inside a loop?

A) Stops the loop entirely B) Skips current iteration and moves to next
C) Restarts the loop from 0 D) Exits the program

65. What is another name for the enhanced `for` loop in Java?

A) Iterator loop B) For-Each loop
C) Index loop D) Dynamic loop

66. Which part of a `for(init; cond; update)` header is optional?

A) Init only B) Cond only
C) Update only D) All parts are optional

67. What creates an infinite loop structure using `for`?

A) `for(;;)` B) `for(true)`
C) `for(inf)` D) `for(0)`

68. What happens if the condition in a `while` loop is initially `false`?

A) Runs once B) Code inside block never executes
C) Compilation Error D) Runtime Error

69. How do labeled `break` statements behave in nested loops?

A) Syntax Error B) Breaks out of the specific labeled outer loop
C) Breaks all loops in program D) Resets iteration counter

70. Can an enhanced `for` loop modify elements of a primitive array directly?

A) Yes
B) No, it uses a local copy of element
C) Only if elements are positive
D) Only with wrapper classes

71. What is the scope of a variable declared inside the initialization block of a `for` loop?

A) Global
B) Entire Class
C) Only within the loop
D) Outer method

72. How many times will `int i=5; while(i>0){ i--; }` iterate?

A) 4 times
B) 5 times
C) 6 times
D) Infinite

73. What syntax ends a `do-while` block?

A) `while(cond)`
B) `while(cond);` (with semicolon)
C) `end while;`
D) `until(cond)`

74. Can a `boolean` variable be used as a loop condition?

A) Yes
B) No
C) Only in do-while
D) Only if constant

75. Which loop structure is best when total iteration count is known beforehand?

A) `while`
B) `do-while`
C) Standard `for`
D) Infinite loop

SECTION 8: SCANNER CLASS & INPUT OPERATIONS (Q76 – Q85)

76. Which package contains the `Scanner` class?

A) `java.lang`
B) `java.io`
C) `java.util`
D) `java.net`

77. How do you instantiate `Scanner` for console input reading?

A) `new Scanner(System.out)`
B) `new Scanner(System.in)`
C) `Scanner.create()`
D) `new Scanner(File)`

78. Which method of `Scanner` reads an entire line of text including spaces?

A) `next()`
B) `nextLine()`
C) `readString()`
D) `getText()`

79. What method reads the next single token/word up to whitespace?

A) `next()`
B) `nextLine()`
C) `nextChar()`
D) `readWord()`

80. Which exception occurs if `nextInt()` receives a input value like `"abc"`?

A) `NumberFormatException`
B) `InputMismatchException`
C) `IOException`
D) `NullPointerException`

81. How do you close a `Scanner` instance named `sc`?

A) `sc.stop()`
B) `sc.close()`
C) `sc.delete()`
D) `sc.finalize()`

82. Which method checks if there is another integer available to read?

A) `hasNextInt()`
B) `checkInt()`
C) `isNextInt()`
D) `existInt()`

83. What happens when `nextLine()` is called immediately after `nextInt()`?

A) Reads next integer
B) Reads leftover newline character `\n`
C) Throws error
D) Clears buffer

84. Does `Scanner` directly support a `nextChar()` method?

A) Yes
B) No, use `next().charAt(0)`
C) Yes, via `readChar()`
D) Only in Java 17+

85. What parameter in `new Scanner(System.in)` represents standard input?

A) `System.in`
B) `System.out`
C) `System.err`
D) `Console`

SECTION 9: CORE LANGUAGE FUNDAMENTALS (Q86 – Q100)

86. Which keyword indicates that a variable belongs to the class rather than an object?

A) `instance`
B) `static`
C) `shared`
D) `native`

87. What is the size of an `int` variable in Java?

A) 2 bytes
B) 4 bytes (32 bits)
C) 8 bytes
D) Depends on OS

88. What is the root class of all classes in Java?

A) `java.lang.Root`
B) `java.lang.Object`
C) `java.lang.Class`
D) `java.lang.Base`

89. Which of the following is NOT a Java keyword?

A) `static`
B) `boolean`
C) `void`
D) `main`

90. What is the standard extension of compiled Java bytecode files?

A) `.java`
B) `.class`
C) `.exe`
D) `.obj`

91. Which component executes Java Bytecode?

A) JDK
B) JVM (Java Virtual Machine)
C) JAVAC
D) IDE

92. What is explicit casting in Java?

A) Automatic conversion
B) Narrowing conversion using `(type)`
C) String conversion
D) Object creation

93. What is the result of `10 / 4` using integer division in Java?

A) `2.5`
B) `2`
C) `3`
D) Compile error

94. What is the result of `10 % 3` in Java?

A) `3`
B) `1`
C) `0`
D) `3.33`

95. What does the `this` keyword represent?

A) Current class object reference
B) Parent class instance
C) Static method reference
D) Global scope

96. Which operator tests if an object is an instance of a specific class?

A) `isinstance`
B) `typeof`
C) `instanceof`
D) `type`

97. Can a class be declared `static` in Java?

A) Top-level classes can
B) Only nested/inner classes can
C) No classes can
D) All classes are static

98. What is the default boolean value of an array element `boolean[] arr = new boolean[1];`?

A) `true`
B) `false`
C) `null`
D) `0`

99. How is a single-line comment written in Java?

A) `# Comment`
B) `// Comment`
C) `<!-- Comment -->`
D) `/* Comment */`

100. Which block always executes after a try-catch block regardless of exceptions?

A) `catch`
B) `final`
C) `finally`
D) `static`

ANSWER KEY (1 – 100)

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	21	B	41	B	61	C	81	B
2	B	22	B	42	C	62	B	82	A
3	B	23	D	43	B	63	C	83	B
4	C	24	B	44	B	64	B	84	B
5	C	25	C	45	B	65	B	85	A
6	B	26	B	46	B	66	D	86	B
7	A	27	A	47	A	67	A	87	B
8	B	28	B	48	B	68	B	88	B
9	B	29	C	49	B	69	B	89	D
10	A	30	B	50	C	70	B	90	B
11	D	31	B	51	B	71	C	91	B
12	B	32	B	52	C	72	B	92	B
13	B	33	C	53	B	73	B	93	B
14	B	34	B	54	B	74	A	94	B
15	B	35	B	55	A	75	C	95	A
16	B	36	B	56	B	76	C	96	C
17	B	37	C	57	B	77	B	97	B
18	B	38	B	58	B	78	B	98	B
19	B	39	B	59	B	79	A	99	B
20	B	40	B	60	B	80	B	100	C